

Lecture Notes on **Core Math**

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Lecture notes on core mathematics. This document now collects the algebra and geometry/topology material in one place. Resources used:

- Algebra:
 - Aluffi's *Algebra: Chapter 0*
 - Atiyah and MacDonald's *Introduction to Commutative Algebra*
- Geometry and topology:
 - Frederic Schuller's *Geometrical Anatomy of Theoretical Physics* video lectures
 - Bott and Tu's *Differential Forms in Algebraic Topology*
 - Littlejohn's *Physics 250* lecture notes

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1 Algebra

1.1 Categorical preliminaries

(fill-in)

1.2 Groups

We review the properties of groups here. First, two high-brow definitions of a group:

- A **group** is a group object in $\mathbf{Set}$.
- A **group** is a one-object groupoid.

These aren't bad definitions, but they're more explicitly useful when they're unpacked a bit.

Definition 1. A **group** is a set with an associative binary operation $m_G: G \rightarrow G$ that has identities and inverses.

We can phrase this by saying that a group has two additional unique maps, $\iota: G \rightarrow G$ by $g \mapsto g^{-1}$ and $\text{id}: G \rightarrow G$ by $g \mapsto g$. This is the more categorical and “morally correct” way of putting this.

Example 1.1 (Canonical group examples)

The most canonical group examples are

- Any field with addition, $\mathbb{Z}$, $\mathbb{R}$, $\mathbb{C}$, $\mathbb{Q}$, etc.. Some of them also work with multiplication.
- $\mathbb{Z}/n\mathbb{Z}$: the integers mod n under addition. $(\mathbb{Z}/n\mathbb{Z})^\times$ are the units mod n .
- D_n . The isometries of a planar solid.
- S_n and A_n . The permutations and even permutations of n objects.
- Matrices over some field under addition or multiplication (if the operation is multiplication then needs invertibility).
- Types of functions under addition, e.g. smooth, periodic, etc..

Groups form a category, $\mathbf{Grp}$. Its maps are **group homomorphisms**, which are maps $\varphi: G \rightarrow G'$ that respect the group structure, $\varphi(ab) = \varphi(a)\varphi(b)$. One way of saying this is that the following diagram commutes:

(put diagram)

We now investigate what other universal constructions exist in $\mathbf{Grp}$.

- **Products:** the product in $\mathbf{Grp}$ is the **direct product group**. This means that for all $\varphi_G: A \rightarrow G$, $\varphi_H: A \rightarrow H$, there exists a unique map $\tilde{\varphi}: A \rightarrow G \times H$ making the diagram

$$\begin{array}{ccc}
 & & G \\
 & \nearrow \varphi_G & \nearrow \pi_G \\
 A & \xrightarrow{\tilde{\varphi}} & G \times H \\
 & \searrow \varphi_H & \searrow \pi_H \\
 & & H
 \end{array}$$

commute. Concretely, taking the binary operations on G and H and defining

$$m_G \times m_H: (G \times H) \times (G \times H) \rightarrow G \times H$$

$$(m_G \times m_H)((g_1, h_1), (g_2, h_2)) \mapsto (m_G((g_1, g_2)), m_H((h_1, h_2))).$$

gives the set $G \times H$ a group structure

$$(g_1, h_1) * (g_2, h_2) = (g_1 g_2, h_1 h_2).$$

- **Coproducts:** there is not a generic coproduct in Grp. (talk about free product and Ab)
- **Quotients** (add)

2 Geometry and Topology

2.1 Manifolds

2.2 Differential constructions

Define Ω^* to be the $\mathbb{R}$ -algebra generated by the symbols $dx^1, \dots, dx^n$ satisfying $dx^i dx^i = 0$ and $dx^i dx^j = -dx^j dx^i$. We define the **vector space of C^∞ differential forms** to be

$$\Omega^*(M) = C^\infty(M) \otimes_{\mathbb{R}} \Omega^*.$$

A **differential k -form** is then a function in $\Omega^*(M)$ that has k dx^i 's in it, i.e. $\omega = \omega_I dx^I$ for some multi-index $I = (i_1, \dots, i_k)$. We can rewrite $\Omega^*(M)$ in terms of these spaces as $\Omega^*(M) = \bigoplus_{k=0}^{\dim M} \Omega^k(M)$; this is just saying that you can partition this space into subspaces based on the degree of the differential form. Note the upper summand: by anti-symmetry, $k \leq \dim M$ (because otherwise the forms would be linearly dependent).

Remark 2.1. There is a clear similarity here between $\Omega^*(M)$ and the Fock space $\mathcal{F} = \bigoplus_{n=0}^{\infty} \mathcal{H}_n$ by analogizing n -forms to n -particle states. The similarity increases further when you consider a fermionic Fock space, where you directly identify fermionic fields $\psi \leftrightarrow dx$.

There are two natural constructions on the space of differential forms: the **wedge product** and the **exterior derivative**. The wedge product is just the natural way of multiplying forms; it's directly given by

$$\omega \wedge \eta = \omega_I \eta_J dx^I dx^J.$$

By definition, we can see that $\omega \wedge \eta = (-1)^{\deg \omega \deg \eta} \eta \wedge \omega$, as we pick up minus signs when we move the forms past each other. The natural derivation on $\Omega^*(M)$ is the **exterior derivative**, $d: \Omega^n(M) \rightarrow \Omega^{n+1}(M)$, defined by

- If $f \in \Omega^0(M)$, then $df = \partial_i f dx^i$.
- If $\omega \in \Omega^k(M)$, then $d\omega = df_I dx^I$.

¹This second condition is actually necessary, as we're dealing with the multiplication of the symbols themselves; we can't just apply condition (1) to $dx^i + dx^j$ and get condition (2) for free.

The two most important properties of the exterior derivative are that it's nilpotent and that it obeys graded Leibniz. That is,

$$d^2 \equiv 0 \qquad d(\omega \wedge \eta) = d\omega \wedge \eta + (-1)^{\deg \omega} \omega \wedge d\eta.$$

Notice that the minus sign in the second term only depends on the degree of the first form in the product. This is because you have to anti-commute the d only past ω , thus moving it past $\deg \omega$ dx 's.

Example 2.1 (Div, grad, curl, and all that)

With the exterior derivative we can regain the definitions of div, grad, and curl extremely easily. On $\mathbb{R}^3$, $\Omega^0(M) \cong C^\infty(\mathbb{R}^3) \cong \Omega^3(M)$, and $\Omega^1(M) \cong \Omega^2(M)$, so we can identify these spaces with one another. Let $f \in \Omega^0(M)$. On functions,

$$df = \partial_x f dx + \partial_y f dy + \partial_z f dz \leftrightarrow \nabla f$$

On one-forms,

$$d(f_i dx^i) = (\partial_x f_y - \partial_y f_x) dx \wedge dy + (\partial_y f_z - \partial_z f_y) dy \wedge dz + (\partial_z f_x - \partial_x f_z) dz \wedge dx \leftrightarrow \nabla \times f$$

And on two-forms,

$$d(2\partial_{[i} f_{j]} dx^i \wedge dx^j) = (\partial_x f_x + \partial_y f_y + \partial_z f_z) dx \wedge dy \wedge dz \leftrightarrow \nabla \cdot f$$

In summary,

$$\begin{aligned} d(0\text{-forms}) &= \text{gradient}, \\ d(1\text{-forms}) &= \text{curl}, \\ d(2\text{-forms}) &= \text{divergence}. \end{aligned}$$

Pretty cool!

(talk about cohomology theory here, then rewrite the first part of this talking about how getting to cohomology is one of our explicit goals)